

Analysis, Design, and Control of the Dual Active Bridge Converter for EV Battery Charging

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Abstract— The growing demand for electric vehicles (EVs) has increased the need for high-quality EV battery chargers. The dual active bridge (DAB) converter, with its safety features like high-frequency galvanic isolation and bidirectional power flow, has become a promising solution for efficient EV battery charging. This paper explores the mathematical modeling, control, and design of the DAB converter-based EV battery charging system under different operating conditions. In order to regulate the power transfer, a single-phase shift (SPS) modulation strategy is employed during constant current (CC) /constant voltage (CV) charging modes. A prototype scaled model rated at 24W is developed in the laboratory environment with operating frequency of 10kHz and tested under varying supply conditions.

Index Terms—Battery charging system, dual active bridge converter, electric vehicle, isolated bi-directional dc-dc converter, single phase shift modulation.

I. INTRODUCTION

With increasing concerns related to environmental issues and excessive dependency on crude oil, electric vehicles (EVs) are gaining popularity among consumers, researchers, and practice engineers to meet these challenges [1]. Nowadays, automobile companies focus on developing electric automotive infrastructure as an alternative to a petroleum-based transportation system [2]. With the development of EV technology, various utility-based electric vehicles are available on the roads, like battery-powered electric vehicles (BEVs), hybrid electric vehicles (HEVs), Plug-in hybrid electric vehicles (PHEVs), and fuel cell-operated vehicles. Given the scarcity of power, these aforesaid vehicles can feed power to meet the local demand. Further, researchers are moving towards grid-interactive EV technology in order to fulfill the demand for power on the consumer side. Electric vehicles can deliver or receive power with the help of bidirectional power electronics converters like DAB converter technology. Additionally, it has a distinctive feature of modular H-bridge configurations, ease of control, galvanic isolation, and soft-switching capability. Therefore, it is preferred for EVs/PHEVs and DC microgrid applications.

Various control algorithms have been suggested in the literature to regulate power transfer through the DAB converter. These include control methods such as single-phase shift (SPS) control, dual-phase shift (DPS) control, and triple-phase shift control (TPS) [3]-[5]. Usually, three parameters, namely phase shift (ϕ) and duty ratios (D_1 , D_2), are responsible for obtaining the required power transfer through the DAB converter. Where ϕ represents the phase shift

between two bridges and D_1 and D_2 , denote the phase displacement within two legs inside the respective bridges [6].

Single-phase (1- ϕ) DAB converter is usually applied for low and medium-power applications. A 1- ϕ DAB converter requires a large filter volume at a high power level. This problem can be resolved by using a three-phase (3- ϕ) DAB converter [7]. The switching losses can be reduced using zero voltage switching (ZVS) based solid-state transformer [8]. The functionality of the DAB converter, which incorporates an extended ZVS range, is explained in detail in [9],[10].

Generally, EV/PHEV chargers are widely classified into two categories: on-board and off-board chargers. Due to the limited EV charging infrastructure, on-board chargers are extensively preferred over off-board chargers. The power capacity of on-board chargers is restricted because of the limited size and space available within EV/PHEV. These problems can be addressed using off-board chargers. The off-board chargers can also interact with the grid and other energy sources. Therefore, the scope of off-board chargers is also increasing in the case of microgrid applications. Further, the power quality issues are the primary concern for the grid-connected off-board chargers. The 3-level DAB converter-based battery charger operated by the SEPIC converter is illustrated in [11].

The performance of EV/PHEVs depends on the battery parameters such as energy storage capacity, lifetime, and cost. To minimize size complexity, the charging infrastructure should be further increased by installing a rapid charging station. As a result, the driving span will improve at less storage capacity [12]-[13]. The DC microgrid integrated charging stations are anticipated to be developed for remote areas. Many charging stations based on solar PV and other distributed energy sources are proliferating. A grid-integrated solar PV micro inverter-based battery charging system is discussed in [14].

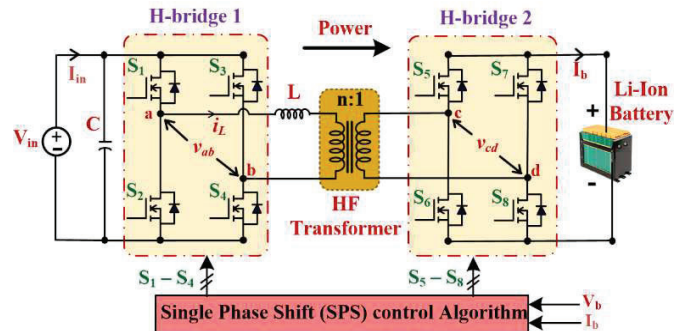


Fig. 1. The Schematic of DAB converter model

This paper aims to develop a single-phase DAB converter-based EV/PHEV charger and test its performance under different charging modes. Further, it can be integrated with solar PV systems. The proposed model is designed for a 24 W power rating and 10 kHz operating frequency. An SPS modulation algorithm has been incorporated to regulate power transmission through the converter. The charging process switches between constant current (CC) and constant voltage (CV) modes based on the battery's state of charge (SOC) to ensure optimal charging conditions.

II. SYSTEM CONFIGURATION AND ITS OPERATION

This section presents details about the DAB converter and its steady-state operation.

A. Topology

The schematic diagram of the single-phase DAB converter-based battery charging system is depicted in Fig 1. It consists of two H-bridge circuits (H bridge-1 and H bridge-2), an inductor (L), and a high-frequency transformer (HFT). Each H-bridge comprises four self-commutating power semiconductor switches (IGBTs/ MOSFETs). The inductor (L) represents the equivalent leakage inductance of the transformer referred to the input side. To compensate for the lower leakage inductance of HFT, an external inductor may be added in series with the inductor (L) to meet the required value of the passive element to feed the power to the battery/load. This element (L) is responsible for power transfer through the converter. Input supply DC voltage and current are represented as V_{in} and I_{in} , respectively. The battery voltage and currents are denoted as V_b and I_b . v_{ab} and v_{cd} are the primary and secondary voltage of HFT.

First input dc supply (V_{in}) is fed to H Bridge-1 of the DAB converter and converted into ac supply (v_{ab}) of the square waveform by switching the switches (S_1, S_4) for half of the switching period (T_s) and other switches (S_2, S_3) for the remaining half period. This voltage (v_{ab}) is applied to the secondary side H Bridge through HFT and the filter inductor. Single-phase HFT provides galvanic isolation and is useful for voltage scaling as per the turn ratio of $n:1$. The secondary side H bridge-2 provides the required DC supply to charge the battery. Regulating power transfer via the converter involves employing an SPS modulation technique that coordinates the operation of two H-bridges.

B. Steady State Operation

Prior to steady state analysis, a few assumptions, like power semiconductor switches and passive elements, are considered ideal. Fig. 2 displays different waveforms of DAB converter parameters, accompanied by their respective switching signals.

Based on the switching states of various switches, four operating modes are identified over one switching cycle, as shown in Fig. 2. These modes are classified based on inductor current variation with a phase shift between two bridges. T_s is the period of one cycle, and T_{sh} denotes half period, i.e., $T_{sh} = T_s/2$. The dT_{sh} is the phase delay between two H bridges,

where parameter 'd' denotes the duty ratio, which serves as the control variable for adjusting the phase difference between two consecutive bridges. Leakage inductance current i_L depends upon the voltage across the inductor, which increases with positive voltage and decreases with negative voltage across the inductor (L). The mathematical analysis of DAB converter in each mode is carried out in the following subsections:

Mode I ($0 \leq t \leq dT_{sh}$)

During this mode, switches S_1 and S_4 of H-bridge-1 and S_6 and S_7 of H-bridge-2 are turned ON, while all other switches in both H-bridges are turned OFF. The current in the inductor rises as the voltage across the inductor is $(V_{in} + nV_b)$, i.e., positive. The equivalent circuits indicating the current path for $i_L < 0$ and $i_L > 0$ are shown in Fig. 3 (a) and Fig. 3(b), respectively. The voltage and current equations at different circuit parts are given below.

$$v_{ab} = V_{in}; v_{cd} = -V_b; v_L = V_{in} + nV_b; i_L = \frac{V_{in} + nV_b}{L}t - I_{Lmin}; \quad (1)$$

Where I_{Lmin} is the initial value of current at the start of mode-1. This mode ends at instant $t = dT_{sh}$.

Mode II ($dT_{sh} \leq t \leq T_{sh}$)

In this mode, The H bridge-1 is activated with switches S_1 and S_4 turned on, while in H bridge-2, switches S_5 and S_8 are on, and all other switches are off. During the dead time between the transition state of switches, ZVS takes place. In this mode, the voltage and current equations are provided below.

$$v_{ab} = V_{in}; v_{cd} = V_b; v_L = V_{in} - nV_b; i_L = \frac{V_{in} - nV_b}{L}(t - dT_{sh}) - i_L(dT_{sh}); \quad (2)$$

During this interval, v_{ab} is positive and equal to V_{in} , and v_{cd} is positive and equal to V_b . The direction of the inductor current is illustrated in Fig 3 (c). The mode-II ends at time instant $t = T_{sh}$.

Mode III ($T_{sh} \leq t \leq T_{sh} + dT_{sh}$)

During this mode, switches (S_5, S_8) remain to conduct while switches (S_3, S_2) are turned ON and remaining switches (S_1, S_4, S_6, S_7) are OFF. The directed circuit during this mode is depicted in Figs. 3(d, e). In this mode, the voltage and current equations are provided below.

$$v_{ab} = -V_{in}; v_{cd} = V_b; v_L = -(V_{in} + nV_b); i_L = -\frac{V_{in} + nV_b}{L}(t - T_{sh}) - i_L(T_{sh}); \quad (3)$$

In this interval, the inductor current decreases as the slope of i_L is negative. Therefore, the conduction path during positive and negative currents is shown in Fig. 3(d) and Fig. 3(e), respectively.

Mode IV ($T_{sh} + dT_{sh} \leq t \leq T_s$)

In mode- IV, the switches (S_2, S_3) remain ON, and switches (S_6, S_7) start conducting. The switches (S_5, S_8, S_1, S_4) are OFF. In this mode, primary voltage (v_{ab}) and secondary voltage (v_{cd}) are negative, and current in inductor, i_L , continues to fall with the slope given in the following i_L equation.

$$v_{ab} = -V_{in}; v_{cd} = -V_b; v_L = -V_{in} + nV_b; i_L = -\frac{V_{in} - nV_b}{L}(t - dT_{sh} - T_{sh}) - i_L(dT_{sh} + T_{sh}); \quad (4)$$

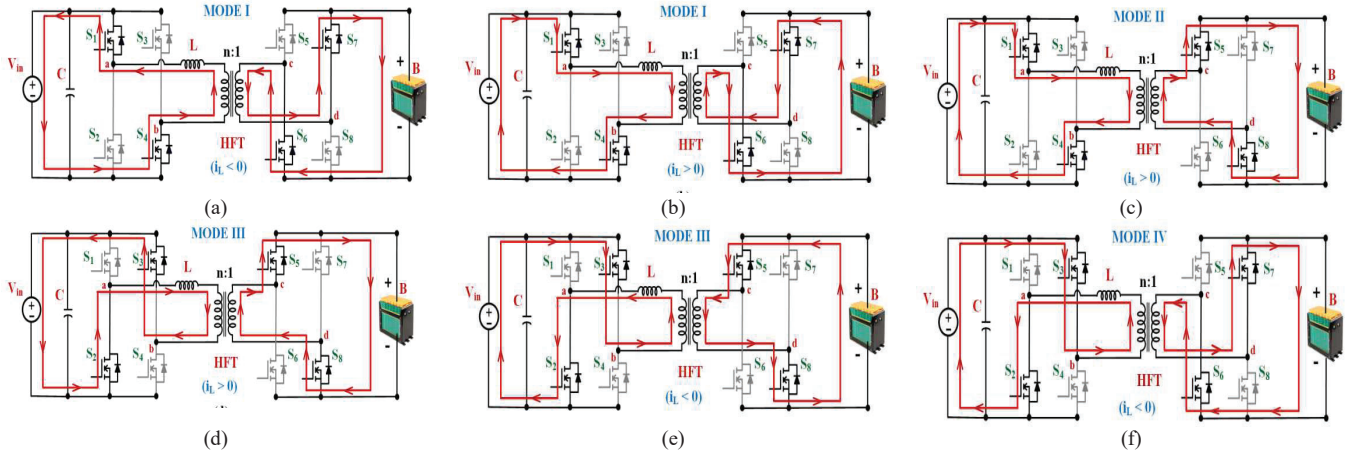


Fig. 3. Mode-wise equivalent circuit indicating current path

C. Power flow analysis

The Fourier analysis is employed for the voltages (v_{ab} , v_{cd}) shown in Fig. 2, and equations (5) (8) are obtained as given below.

$$v_{ab}(t) = \sum_{k=1,3,5,\dots} \frac{4V_{in}}{k\pi} \sin(k\omega t) \quad (5)$$

The fundamental component of voltage $v_{ab}(t)$,

$$v_{ab1}(t) = \frac{4V_{in}}{\pi} \sin(\omega t) \quad (6)$$

The R.M.S. value of $v_{ab}(t)$, is obtained from equation (6).

$$V_{ab1} = \frac{4V_{in}}{\sqrt{2}\pi} = \frac{2\sqrt{2}}{\pi} V_{in} \quad (7)$$

$$v_{cd}(t) = \sum_{k=1,3,5,\dots} \frac{4V_b}{k\pi} \sin(k\omega t - \phi) \quad (8)$$

The fundamental component of voltage $v_{cd}(t)$ is obtained in equation (9).

$$v_{cd1}(t) = \frac{4V_b}{\pi} \sin(\omega t - \phi) \quad (9)$$

The R.M.S value of the voltage $v_{cd}(t)$ is obtained from equation (9).

$$V_{cd1} = \frac{4V_b}{\sqrt{2}\pi} = \frac{2\sqrt{2}}{\pi} V_b \quad (10)$$

The simplified circuit representation of the DAB converter and its phasor diagram illustrating fundamental quantities are depicted in Fig. 4. The primary voltage v_{ab1} is taken as reference, and voltage (nV_{cd1}) referred to the primary side of HFT is lagging behind v_{ab1} by phase shift angle of ϕ .

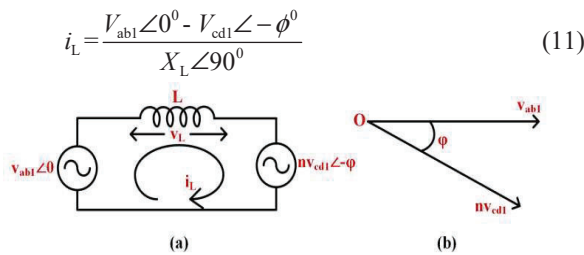


Fig. 4. Approximate equivalent circuit and its phasor diagram.

The complex power is transferred to output through the inductor (L),

$$S_1 = P_1 + jQ_1 = nV_{cd1} \angle -\phi \times i_L^* \quad (12)$$

$$S_1 = nV_{cd1} \angle -\phi \left(\frac{V_{ab1} \angle 0^\circ - nV_{cd1} \angle -\phi}{X_L \angle 90^\circ} \right)^*, \text{ where } X_L = 2\pi fL$$

$$S_1 = \frac{nV_{ab1}V_{cd1}}{X_L} \angle (-\phi + 90^\circ) - \frac{n^2V_{cd1}^2}{X_L} \angle 90^\circ \quad (13)$$

The active power transferred to load is given below.

$$P_1 = \frac{nV_{ab1}V_{cd1}}{X_L} \cos(-\phi + 90^\circ) - \frac{n^2V_{cd1}^2}{X_L} \cos 90^\circ$$

$$P_1 = \frac{nV_{ab1}V_{cd1}}{X_L} \sin \phi \quad (14)$$

From equation (7) and (10), the power (P_1) is altered as given in equation (15)

$$P_1 = \frac{0.129nV_{in}V_b}{fL} \sin \phi \quad (15)$$

The maximum power transferred at the output is obtained as given below:

$$\frac{dP_1}{d\phi} = \frac{d}{d\phi} \left(\frac{0.129nV_{in}V_b}{fL} \sin \phi \right) = 0$$

$$\cos \phi = 0, \quad \phi = 90^\circ = 0.5T_{sh}$$

Hence, maximum power is transferred at 90° phase shifts with 0.5 duty ratio. Therefore, the maximum power ($P_{1\max}$),

$$P_{1\max} = \frac{0.129nV_{in}V_b}{fL} \quad (16)$$

C. Design Consideration

In this subsection, the design of the inductor (L) of the DAB converter is carried out. The value of the inductor corresponding to the maximum power transferred to output can be evaluated from equation (16) as given below.

$$L = \frac{0.129nV_{in}V_b}{fP_{1\max}} \quad (17)$$

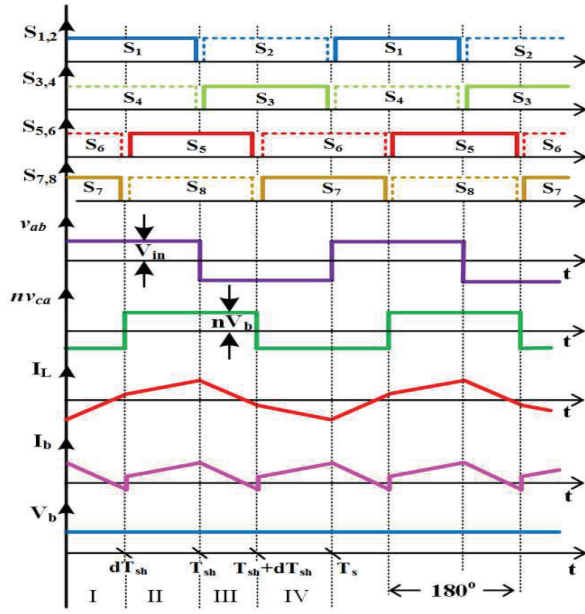


Fig. 2. Steady state switching waveforms of voltage and current.

$$n = 0.5, V_{in} = 24V, V_b = 12V, f = 10 \text{ kHz}, P_{lmax} = 24W$$

$$L = \frac{0.129 \times 0.5 \times 24 \times 12}{10 \times 10^3 \times 24} = 7.74 \times 10^{-5} H = 77.4 \mu H$$

Therefore, 77.5 μH inductance (including leakage inductance of HFT) is required for 24W power rating.

III CONTROL STRATEGY

The proposed DAB converter model utilizes an SPS control algorithm for regulating the energy transfer to the EV battery. In order to charge the battery, the H-bridge 2 is operated at phase delay of ϕ lagging behind the H-bridge 1 by phase shift modulation technique. The DAB converter functions in two different modes, i.e., constant current (CC) mode and constant voltage (CV) mode. The mode choice depends on the battery's SOC. The block representation of the suggested control algorithm is depicted in Fig. 5.

The current controller operates in CC mode, and the voltage controller operates in CV mode. The mode selector block selects the operating mode by observing the SOC of the battery. When the SOC of the battery is less than 80%, the battery will charge in CC mode. The system compares the desired battery current $I_b^*(k)$ with the actual sensed battery current $I_b(k)$ at a specific moment k . The resulting disparity between the desired and sensed currents produces an error signal $I_e(k)$. This error signal is pivotal in control

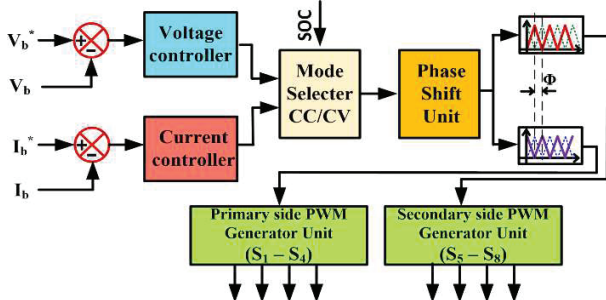


Fig. 5. Control schematic of battery charging during CC and CV mode.

systems, serving as feedback. It guides adjustments to the system, ensuring that the sensed battery current aligns closely with the desired value. The control signal $I_c(k)$ is determined by equation (19), where K_{pc} and K_{ic} represent the proportional and integral gains of the current controller, respectively.

$$I_e(k) = I_b^*(k) - I_b(k) \quad (18)$$

$$I_c(k) = I_c(k-1) + K_{pc} \{I_e(k) - I_e(k-1)\} + K_{ic} I_e(k) \quad (19)$$

When the SOC of the battery reaches above 80%, the charging of the battery will be shifted to CV mode using the mode selector block. At any instant k , the battery voltage $V_b(k)$ is compared with reference voltage $V_b^*(k)$ yield error signal $V_e(k)$ fed to the voltage controller, as shown in Fig 5. The voltage controller produces the control voltage $V_c(k)$ to maintain the battery voltage constant. The expression of $V_c(k)$ is given in equation (21), where K_{pv} and K_{iv} are the proportional gains of the voltage controller.

$$V_e(k) = V_b^*(k) - V_b(k) \quad (20)$$

$$V_c(k) = V_c(k-1) + K_{pv} \{V_e(k) - V_e(k-1)\} + K_{iv} V_e(k) \quad (21)$$

Control signals are compared with 10 kHz carrier signals to create switching signals for switches (S_1 - S_8) and regulate power transfer to the battery.

IV. RESULT AND DISCUSSION

A 24W, 12 V DAB converter prototype is developed for battery charging applications. 24V input dc supply and 10 kHz switching frequency are selected to operate the DAB converter. A laboratory prototype with a low power rating is created to confirm the accuracy of the simulation results. The outcomes of both the simulation and the physical tests are outlined below.

A. Simulation results

A Simulink model of the DAB converter is developed with the help of the MATLAB toolbox, and its simulation study is carried out for charging EV batteries. The simulation study focused on charging an EV battery, explicitly examining the charging process under CC and CV modes. The simulation observes key converter circuit parameters, as depicted in Fig. 6, during a steady-state condition. The waveforms of voltages and currents (v_{ab} , v'_{cd} , v_L , i_L) are shown in Fig. 6 (a). It has been observed that phase shift of ϕ between v_{ab} and v'_{cd} appeared which gives voltage across the inductor (v_L). Further, as per the v_L , the inductor current (i_L) increases for positive voltage and decreases for negative voltage as seen in Fig.6(a). Fig. 6(b) depicts voltage and current information for both sides of the HFT. In contrast, Fig. 8(c) shows the input source voltage, current, as well as battery voltage and current for the DAB converter, i.e., v_{in} , i_{in} , V_b , and i_b . The HFT with turn ratio 2:1 is selected to scale down the voltage suitable for battery. These results are observed at 30% SOC of the battery. It has been observed that 2 A constant dc current is appeared across the battery for charging. Fig. 6 (d) shows the voltage and current waveforms associated switch S_1 and S_2 . It is observed that ZVS for the all switches of the DAB converter is achieved.

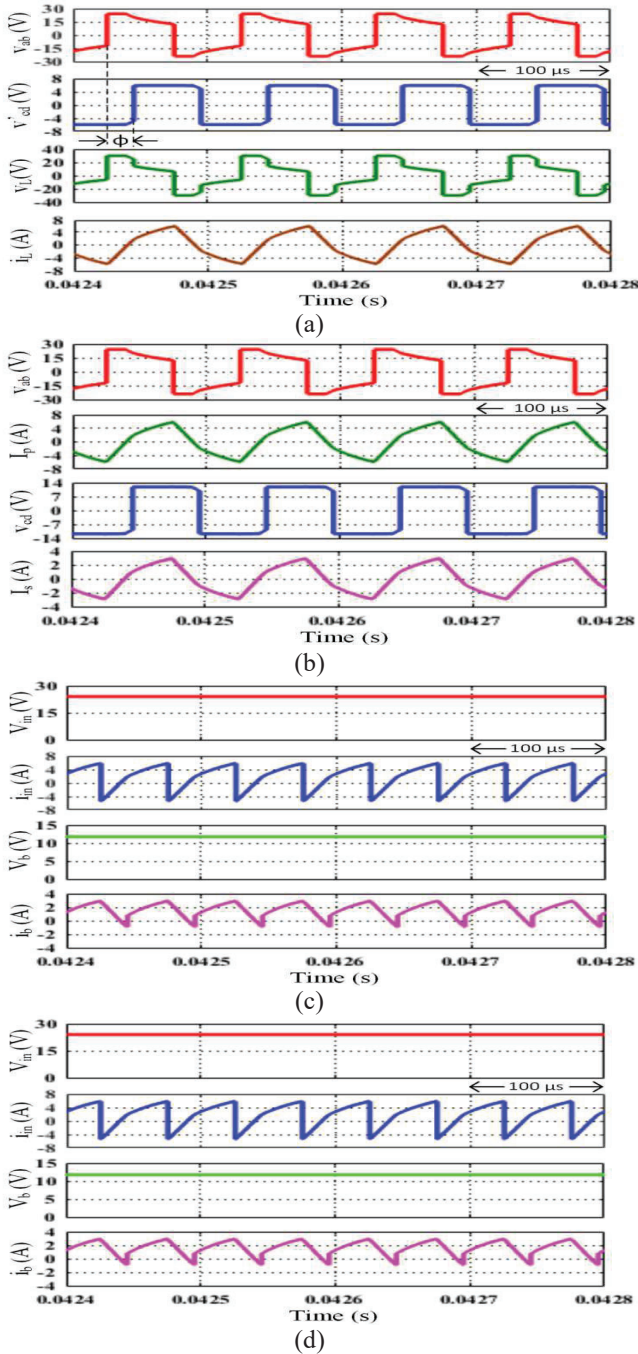


Fig. 8. Simulation results of DAB converter feeding battery under PS control (a) v_{ab} , v_{cd}' , v_L , i_L (b) v_{ab} , i_p , v_{cd} , v_L , i_s (c) V_m , i_{in} , V_b , i_b (d) Switching signals (S_1 , S_2 v_{SI} , i_{SI}),

During the activation of switch S_1 , it has been noticed that there is no voltage across the switch, as illustrated in Fig. 6 (d).

The effect of voltage variation on battery current is illustrated in Fig. 7. It has been observed that the slight variation in input voltage changes the ripple current across the battery, but due to the action of the current controller, the average value of the battery current remains the same as previous. The effect of increasing and decreasing input voltage is illustrated in Fig. 7 (a) and Fig. 7 (b), respectively.

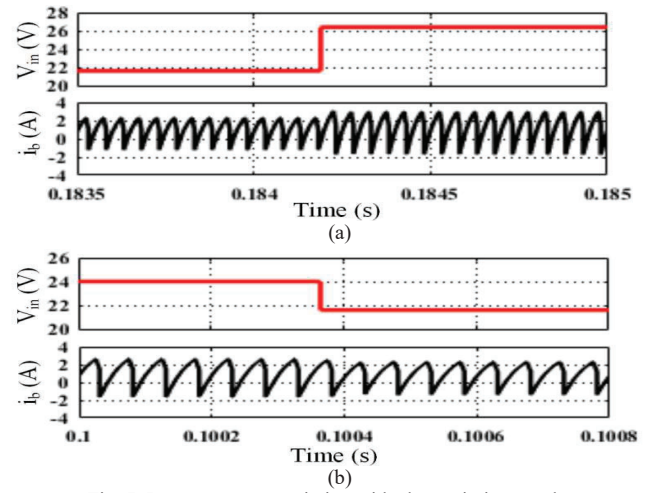


Fig. 7. Battery current variation with change in input voltage

B. Test results

A preliminary version of the DAB converter has been created in the lab setting to confirm the effectiveness of the phase-shift control algorithm for battery charging application, depicted in Fig. 10. The single-phase AC supply generates an input DC supply of 24V for the DAB converter, using a single-phase autotransformer, a diode bridge rectifier (DBR), and a capacitor ($2 \times 470 \mu\text{F}$, 400V). The power circuit of the DAB converter is developed using eight units of IRFP540 along with their driver circuits, high-frequency transformer (HFT), and inductor. The STM32F407VG microcontroller is used to implement phase-shift control, and necessary switching signals are generated to operate two H-bridge inverters of DAB. The system's performance has been investigated by utilizing results obtained through a four-channel digital storage oscilloscope (DSO) and differential probe made by Agilent. To assess the DAB converter's performance under SPS control and battery charging in CV mode, Fig. 11 illustrates the steady-state voltage and current waveforms at different test points for investigation. Fig.11 (a) shows the waveform of voltages (v_{ab} , v_{cd}' , v_L , i_L). It is observed that waveforms of v_{ab} and v_{cd}' are phase-shifted, yielding v_L and i_L , as shown in Fig.11 (a). Fig.11(b) depicts waveforms of voltages (v_{ab} , v_{cd}) and current (i_p , i_s) of high-frequency transformers. The waveforms of input voltage (V_m), input current (i_{in}), battery voltage (V_b), and battery current (i_b) are depicted in Fig.11(c).

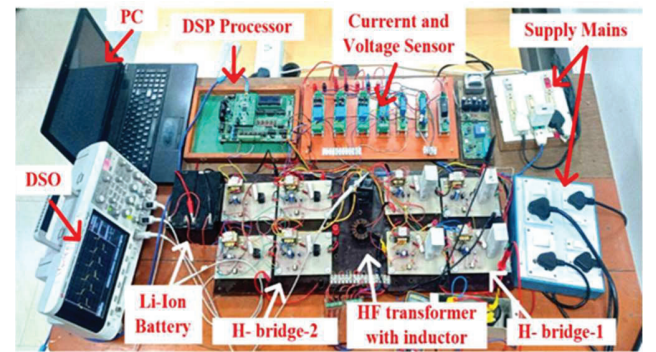


Fig. 10. Photograph of test setup of DAB converter feeding battery.

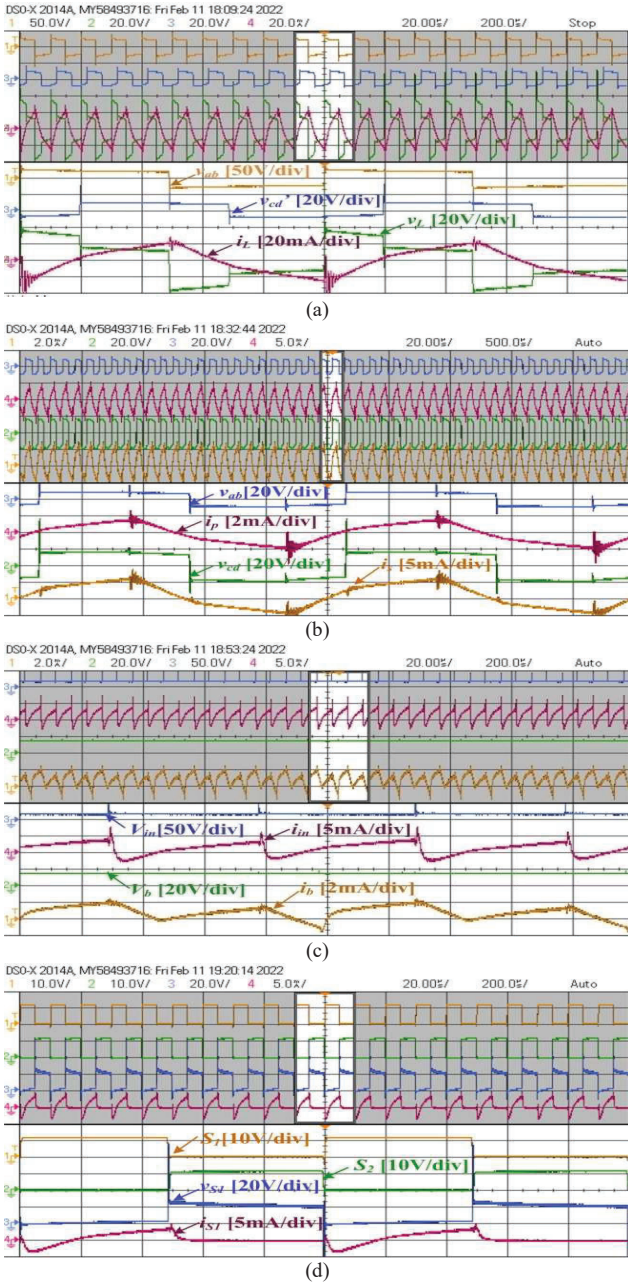


Fig.11. Steady-state waveforms of DAB converter feeding battery under PS control (a) v_{ab} , v_{cd}' , v_L , i_L (b) v_{ab} , i_p , v_{cd} , v_L , i_s (c) V_{in} , i_{in} , V_b , i_b (d) Switching signals (S_1 , S_2), v_{S1} , i_{S1}

In order to investigate the features of ZVS of power semiconductor switches (S_1 to S_8) of H-bridge inverters, the switching signals of switches (S_1 , S_2) and the voltage across switch S_1 i.e., v_{S1} and i_{S1} are recorded as shown in Fig.11(d). It is observed that before switching off S_1 , its current is negative. It ensures that the body diode of switch S_1 conducts prior to transitioning to S_1 . Similar waveforms are also obtained for the remaining switches.

V. CONCLUSION

In this paper, the performance of a DAB converter employed for battery charging is investigated through simulation, and later on, it is validated through

experimentation on a prototype model. Detailed mathematical modeling under different modes is carried out. Further, the power flow analysis is presented. The SPS modulation technique regulate power supply for charging electric vehicle batteries in CC and CV modes. The performances of ZVS-enabled DAB under steady state and dynamic conditions are investigated through the Simulink model. The simulation outcomes are validated through a hardware prototype developed in the laboratory environment. The developed setup is tested under various SOC's of the battery and different input DC supply voltages. It is observed that the test setup gives a stable response under steady-state conditions.

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